

第三次计算机组成原理作业

3.14. (1) $[X]_{\text{补}} = 0.11001$ $[Y]_{\text{补}} = 1.01001$

$$[X]_{\text{补}} + [Y]_{\text{补}} = 0.00010$$

(2) $[X]_{\text{补}} = 0.10010$ $[Y]_{\text{补}} = \overset{0.11000}{\text{1.01110}}$

$$[X]_{\text{补}} + [Y]_{\text{补}} = \text{0.00000} \quad 1.01010 \text{ 溢出}$$

3.15. (1) $[X-Y]_{\text{补}} = [X]_{\text{补}} + [-Y]_{\text{补}}$

$$[X]_{\text{补}} = 1.10001 \quad [-Y]_{\text{补}} = 1.11011$$

$$[X]_{\text{补}} + [-Y]_{\text{补}} = 1.01100$$

(2) $[X]_{\text{补}} = 0.11011$ $[-Y]_{\text{补}} = 0.10010$ 溢出

$$[X-Y]_{\text{补}} = [X]_{\text{补}} + [-Y]_{\text{补}} = 1.01101$$

3.16. (1) $[X]_{\text{补}} = 11101111$ $[Y]_{\text{补}} = \overset{00101011}{\text{01010101}}$

$$[X+Y]_{\text{补}} = \overset{00011010}{\text{00011010}} \quad [X+Y]_{\text{移}} = 10011010$$

(2) $[X]_{\text{补}} = 01101111$ $[Y]_{\text{补}} = 11010101$

$$[-Y]_{\text{补}} = 00101011$$

$$[X+Y]_{\text{补}} = 01000100 \quad [X+Y]_{\text{移}} = 11001000$$

$$[X-Y]_{\text{补}} = 10011010 \quad [X-Y]_{\text{移}} = 00011010$$



3.17

$$\begin{array}{r}
 0.1101 \\
 \times 0.1011 \\
 \hline
 01101 \\
 01101 \\
 01101 \\
 \hline
 0.10001111
 \end{array}$$

加上符号有 1.10001111

3.18. $[X]_{\text{补}} = 0.1010$, $[Y]_{\text{补}} = 1.1010$

$$[X \cdot Y]_{\text{补}} =$$

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$y_i y_{i+1}$	部分积	乘积位
00	00.0000	1.1010 0
00	00.0000	0.11010
10 + $[-X]_{\text{补}}$	01.0110	0.01101
01 + $[X]_{\text{补}}$	00.1011	
	01.0001	0.00110
	00.1000	
10 + $[-X]_{\text{补}}$	11.1110	0.00011
	0 1.1111	
11	11.1111	

因此答案为 11.1111

